

Workshop Sheet - Week 2

Sets, functions and relations in Lean. Below are some of the new tactics we will be using today.

<code>constructor</code>	Split an $\wedge$ goal or prove both directions of $\leftrightarrow$.
<code>ext x</code>	Prove equality by comparing membership or values at x .
<code>change P</code>	Replace the goal by a definitionally equal expression P .
<code>calc</code>	Write a chain of equalities, giving a proof for each step.
<code>ring</code>	Prove a polynomial identity by expanding and collecting terms.

Remarks. The notation `fun x => x + 1` defines the function $x \mapsto x+1$. For $f : X \rightarrow Y$, $S : \text{Set } X$ and $U : \text{Set } Y$, the expression `f ⁻¹ ' U` denotes the preimage $f^{-1}(U)$ and `f '' S` the image $f(S)$.

1. *Set theory.* Let $S, T \subseteq X$, $U \subseteq Y$, $f : X \rightarrow Y$, and $x \in X$.
 - (a) If $x \in S$ and $x \in T$, prove that $x \in S \cap T$.
 - (b) If $x \in S \cap T$, prove that $x \in S$. (Hint: If $h : P \wedge Q$, then `h.left` is a proof of P).
 - (c) If $S \subseteq T$ and $x \in S$, prove that $x \in T$.
 - (d) If $f(x) \in U$, prove that $x \in f^{-1}(U)$.
 - (e) If $x \in S$, prove that $f(x) \in f(S)$.
 - (f) Prove that $f(S \cap T) \subseteq f(S) \cap f(T)$.
 - (g) Prove that $f(S \cap f^{-1}(U)) = f(S) \cap U$.
 - (h) Prove that $\mathcal{P}(S \cap T) = \mathcal{P}(S) \cap \mathcal{P}(T)$.
2. *Functions.* Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.
 - (a) If $g \circ f$ is injective, prove that f is injective. (Hint: use `change` on $(g \circ f)(x) = (g \circ f)(y)$).
 - (b) If $g \circ f$ is surjective, prove that g is surjective.
 - (c) If $r, s : Y \rightarrow X$ satisfy $r \circ f = \text{id}_X$ and $f \circ s = \text{id}_Y$, prove that $r = s$.
 - (d) Prove there is an injection $X \rightarrow \mathcal{P}(X)$ given by $x \mapsto \{x\}$.
 - (e) Prove and formalize the following theorem.

Theorem (Cantor). *For every set X , there is no surjection $X \rightarrow \mathcal{P}(X)$.*
3. *Relations.* Fix $n \in \mathbb{Z}$ and define $a \sim_n b$ if $n \mid (b - a)$. Write $[a]_n = \{x \in \mathbb{Z} \mid a \sim_n x\}$.
 - (a) Prove that $\sim_n$ is reflexive. (Hint: use `change`).
 - (b) Prove that $\sim_n$ is symmetric. (Hint: use `change` and `calc`).
 - (c) Prove that $\sim_n$ is transitive.
 - (d) Prove that $\sim_n$ is an equivalence relation.
 - (e) If $a \sim_n a'$ and $b \sim_n b'$, prove that $a + b \sim_n a' + b'$.
 - (f) Prove that $[a]_n = [b]_n$ if and only if $a \sim_n b$.